

# Impact Statement

## BRIDGING THE TOKENIZATION GAP IN AFRICAN LLMs

**Project:** SuperBPE for African-Language Tokenization

**Team:** TRI AI Cohort 10 — Team Krugger

An African-language SuperBPE tokenizer for more efficient and inclusive language AI.

### THE PROBLEM

Many African languages are inefficiently tokenized because existing tokenizers are predominantly optimized using high-resource languages such as English.

For languages such as **Igbo, Yoruba, Hausa and Swahili**, a single word or expression can be fragmented into multiple tokens. This consumes valuable context space and can increase the computational resources required to process African-language text.

**The result:** African languages remain more expensive and less efficiently represented in modern AI systems.

### OUR SOLUTION

We extended the **SuperBPE** approach to better capture recurring patterns in African-language text.

Our tokenizer:

**Learns ordinary subword units → learns frequent cross-space expressions → produces more efficient representations of African-language text.**

We also introduced **EncodeLang**, a proposed community-driven layer where native speakers can contribute and validate translations, helping improve evaluation and community ownership of African-language resources.

## WHAT WE BUILT

| Metric                             | Result               |
|------------------------------------|----------------------|
| Vocabulary size                    | <b>19,000 tokens</b> |
| Learned cross-space tokens         | <b>12,345</b>        |
| Tokens for 50,000 characters       | <b>20,715</b>        |
| Held-out characters/token          | <b>6.829</b>         |
| CodaBench runtime                  | <b>0.18 seconds</b>  |
| Local 2M-character runtime         | <b>1.475 seconds</b> |
| Languages in reported training run | <b>4</b>             |

**Languages:** Igbo • Yoruba • Hausa • Swahili

## WHO BENEFITS?

### **African-language speakers**

Better representation of their languages in AI systems.

### **AI researchers**

A foundation for studying efficient African-language tokenization.

### **LLM developers**

More efficient language representations for African-language applications.

### **African businesses & startups**

Potentially more practical and cost-efficient language AI applications.

### **Educators & language organizations**

Better tools and resources for African-language technology.

### **Future generations**

Greater digital representation of African languages and linguistic heritage.

## EXPECTED IMPACT

### 01 — Better representation

Reduce unnecessary fragmentation of African-language text.

### 02 — Greater efficiency

Improve context utilization by representing recurring linguistic patterns more efficiently.

### 03 — More inclusive AI

Help African languages participate more effectively in the LLM ecosystem.

### 04 — Community ownership

Give native speakers a role in validating the language resources used to evaluate AI.

### 05 — Lower computational overhead

More efficient tokenization can reduce the amount of text representation an AI system needs to process.

## HOW WE WILL MEASURE IMPACT

We will track:

- **Token reduction** against relevant baseline tokenizers
- **Characters per token**
- **Tokens per sentence**
- **Lossless reconstruction accuracy**
- **Performance across individual African languages**
- **Native-speaker validation results**
- **Number and diversity of community contributions through EncodeLang**

## OUR LONG-TERM VISION

**African languages should not have to become computationally expensive for Africans to participate in the AI economy.**

We aim to contribute to an AI ecosystem where African languages are efficiently represented, community-informed, and treated as first-class languages in language technology.